NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science & Engineering

MID SEMESTER EXAMINATION, Sept. 2025

B. Tech: CSE, Semester-V

Course Name: Information Security **Code: CS54119**

Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. ***Attempt all questions.***
2. ***Assume any suitable data, if necessary.***
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.*

	Questions	Marks	CO	BL
1.	**a.** Highlights the importance of following security approaches and also mention the example of security mechanism used in each case: Attack Deterrence, Attack Prevention, Attack Deflection, and Attack Avoidance.	3+3	CO2	U,A
	b. Define the type of security attacks in each of the following cases: – A student breaks into a professor's office to obtain a copy of the next day's test. – A student gives a check for \$10 to buy a used book. Later she finds that the check was cashed for \$ 100. – A student sends hundreds of e-mail per day to another student using a phony return e-mail address.		CO1	U,P
2.	**a.** Write down the pseudocode of square-and-multiply algorithm. Find the result of (320^{23} mod 461) using the square-and-multiply algorithm.	2+4	CO1	U, E
	b. Difference between Monoalphabetic Cipher and Polyalphabetic Cipher? Which one is more secure? Justify your answer. Encrypt the following plaintext using Vigenere Cipher. The keyword is "CIPHER". Plaintext: "thiscryptosystemisnotsecure". You should also justify that how Vigenere Cipher is different from Shift Cipher?		CO4	A, P
3.	For Deffie-Hellman key exchange scheme compute the shared secret key using the following chosen parameters: prime=353, primitive root=3, Number chosen by A=97, by B=233. How Diffie- Hellman key exchange scheme is vulnerable to man-in-the-middle attack? What kind of solution may be given to defeat this kind of attack?	6	CO 4	U, P
4	Encrypt the plaintext message NO WAY using the RSA algorithm with key (e, n)= (47, 1537). Use the extended Euclidean algorithm to computer recovery exponent for the cryptosystem? **Hint:** Each number of the plaintext is replaced by a two-digit integer (i.e. A=00, B=01…Z=25) and space is replaced by 99.	6	CO4	A,P
5	**a.** Explain the block diagram of DES algorithm. **b.** A 4-round toy Feistel cipher operates on 8-bit blocks (split 4-bit L_0R_0). Round function: F(R,K) = (left-rotate R by 1 bit) ⊕ K. Given: L_0R_0 = 1011 0010, round keys $K_1…K_4$ = 1100, 0110, 1010, 0101. Compute L_4R_4 after exactly 4 rounds; show each round's working.	2+4	CO1, CO4	P,C

****End of Questions****

Level	Bloom's Taxonomy
1	Remembering (R)
2	Understanding (U)
3	Applying (P)
4	Analyzing (A)
5	Evaluate (E)
6	Create (C)

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END SEMESTER EXAMINATION, (Jul-Dec 2025)

B. Tech: CSE, Semester-V

Course Name: Information Security **Code: CS54119**

Maximum Time: 3 hours **Max. Marks: 60**

Instruction:

1. ***Attempt all questions.***
2. ***Assume any suitable data, if necessary.***
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.*

	Questions	Marks	CO	BL
1.	**a.** Distinguish between a substitution cipher and a transposition cipher. Encrypt "LASTQUESTION" using the playfair cipher and the keyword "LEMONS".	5+5	CO4	Apply
	b. In the Diffie-Hellman protocol, what happens if x and y have the same value, that is, Alice and Bob have accidentally chosen the same private number? Are R1 and R2 the same? Do the session keys calculated by Alice and Bob have the same value? Use an example to prove your claims.		CO4	Analyze Remember
2.	In Password-based authentication, the system keeps a table (file) of two columns: User ID and password in plaintext. Users who want to access the system send their user ID and password in plaintext. If the password stored in the table matches the one provided by the user, access is granted. Is this kind of authentication mechanism secure? If not, suggest changes to above method to enhance the security of authentication. Also, suggest how to prevent dictionary attacks. Your answer should include a detailed pictorial presentation.	10	CO5, CO3	Evaluate Create
3.	**a.** Show the process of Information Risk Management also detail the steps of risk management process.	4+6	CO3	Understand
	b. Approximately 100 users in the organisation use laptop computers. The value of each computer is $1,700, which includes the cost of the computer, the software, and the data. In the past two years, the organization has lost an average of seven computers per year. With this information, calculate the single loss expectancy (SLE), the Annualized rate of occurrence (ARO), and the annualized loss expectancy (ALE.). As a countermeasure to protect against the loss of these computers the locks are purchased in bulk, they cost $15 each, and their use is estimated to reduce yearly losses from seven to only one. Is this a cost-effective countermeasure?		CO3	Apply Evaluate
4	Biba is a model focused on integrity and Bell-LaPadula is focused on confidentiality. These models describe the information flow between different subjects and objects, as well as the use of access controls and permissions to protect confidentiality and integrity. Show with respective diagrams, what security classifications and rules are applicable to demonstrate the applicability of these access control models?	10	CO2, CO5	Understand Analyse
5	**a.** Compare the advantages of Public key Cryptography over Symmetric Key Cryptography and conversely.	5+5	CO4	Understand Analyse
	b. The following three major threat types directly threaten each of the CIA tenets: Disclosure threats, Alteration threats, and Denial or destruction threats. Present a scenario for the applicability of these threats, either separately or in combination.		CO1, CO2	
6	Given is a Feistel Cipher with the following parameters: 8 bit blocks, Ki=1010, F(Ri-1,Ki) = 1111 = const The output of F is always 1111, no matter what the input. This does not make sense for a real cipher, but it allows us to demonstrate some characteristics of Feistel ciphers. Plaintext block: 1001 1100 Calculate the ciphertext using two rounds. Decrypt the ciphertext by applying two Feistel rounds to the ciphertext. You should get the original plaintext. Remember, at the end of each encryption and decryption process, the left and the right half are swapped one more time.	10	CO4	Apply Analyse

****End of Questions****

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